Transformation of Random Variables

L12_S1_A.pdf
PDF

Generate a lecture scribe intended to serve as exam-oriented reference material
for Lecture 12 of CSE 400: Fundamentals of Probability in Computing,
titled “Transformation of Random Variables”.

Use ONLY the provided lecture slides and relevant textbook content as context.
Do not introduce any definitions, examples, derivations, notation, or results
that are not explicitly present in the provided material.

The scribe must include, exactly as presented in the lecture:

• The transformation of a single random variable Y = g(X)
• Derivation of the CDF-based method for finding f_Y(y)
• Treatment of monotonic increasing and monotonic decreasing transformations
• Change of limits and inversion steps
• The worked example where X is uniformly distributed and
  Y = sin(πX/2), including all intermediate derivation steps

• Transformation involving two random variables
• The case Z = X + Y
• Derivation of F_Z(z) using joint PDF integration
• Application of differentiation to obtain f_Z(z)
• Use of Leibniz rule as shown in the lecture
• The resulting convolution formula
• The special case when X and Y are independent
• The convolution expression for independent random variables

• The case where X and Y are Normal random variables

and the
derivation showing that Z is also Normal
• The case where X and Y are Exponential random variables and
derivation of f_Z(z)
• The transformation Z = X − Y
• The transformation Z = X / Y with appropriate region analysis
• The example involving R = sqrt(X^2 + Y^2), as presented

Organize the scribe following the logical structure and progression of
the lecture. Clearly state all definitions, assumptions, limits of integration,
region descriptions, and intermediate algebraic steps exactly as shown in the
context.

The scribe must be complete, rigorous, and suitable for revision for a
closed-notes examination. Do not simplify, shorten, or omit derivations.
Do not add intuition or commentary beyond what appears in the lecture.
Do not generate new examples.

Below is the complete lecture scribe for **Lecture 12: Transformation of Random Variables**, prepared strictly from the provided lecture slides . All definitions, derivations, limits, and intermediate steps follow exactly the structure and development shown in the lecture.

CSE 400 – Fundamentals of Probability in Computing

Lecture L11_12: Transformation of Random Variables

(Source: Lecture Slides)

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1. Transformation of a Single Random Variable

Let X be a continuous random variable with known PDF $f_X(x)$ and CDF $F_X(x)$.

Define a new random variable:

$$Y = g(X)$$

Goal: Determine $F_Y(y)$ and $f_Y(y)$.

Step 1: CDF of Y

$$F_Y(y) = P(Y \leqslant y)$$
$$= P(g(X) \leqslant y)$$

The derivation depends on whether $g(\cdot)$ is monotonic increasing or monotonic decreasing.

Case 1: $g(\cdot)$ Monotonically Increasing

Since g is increasing,

$$g(X) \leqslant y \Leftrightarrow X \leqslant g^{-1}(y)$$

Thus,

$$F_Y(y) = P(X \leqslant g^{-1}(y))$$
$$= F_X(g^{-1}(y))$$

Differentiate w.r.t. y:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Since

$$\frac{d}{dy}g^{-1}(y) = \frac{dx}{dy}\bigg|_{x=g^{-1}(y)}$$

Hence,

$$f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right|_{x=g^{-1}(y)}$$

Case 2: $g(\cdot)$ Monotonically Decreasing

Now,

$$g(X) \leqslant y \;\Leftrightarrow\; X \geqslant g^{-1}(y)$$

Thus,

$$F_Y(y) = P(X \geqslant g^{-1}(y))$$
$$= 1 - F_X(g^{-1}(y))$$

Differentiate:

$$f_Y(y) = -f_X(g^{-1}(y))\frac{d}{dy}g^{-1}(y)$$

Since derivative is negative, taking magnitude:

$$f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right|_{x=g^{-1}(y)}$$

Change of Limits and Inversion

When solving:

1. Express $y = g(x)$
2. Invert to get $x = g^{-1}(y)$
3. Compute $\frac{dx}{dy}$
4. Change support using transformed endpoints

Worked Example

Given:

$$X \sim \text{Uniform}(-1, 1)$$

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Define:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Step 1: Invert Transformation

$$y = \sin\left(\frac{\pi x}{2}\right)$$

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Step 2: Derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1 - y^2}}$$

Step 3: Support

When $x = -1$,

$$y = \sin(-\pi/2) = -1$$

When $x = 1$,

$$y = \sin(\pi/2) = 1$$

Thus,

$$-1 < y < 1$$

Step 4: Compute PDF

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

$$= \frac{1}{2} \cdot \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

$$= \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1$$

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2. Transformation of Two Random Variables

Let

$$Z = X + Y$$

Find:

1. $f_Z(z)$
2. Case when X, Y independent
3. Normal case
4. Exponential case

CDF of Z

$$F_Z(z) = P(Z \leqslant z)$$

$$= P(X + Y \leqslant z)$$

$$= \iint_{x+y \leqslant z} f_{XY}(x, y)\, dx\, dy$$

From the geometric region shown in the lecture:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{XY}(x, y)\, dx\, dy$$

Differentiation to Obtain PDF

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Leibniz Rule (as given)

If

$$G(z) = \int_{a(z)}^{b(z)} h(z, y)dy$$

then

$$\frac{d}{dz}G(z) = \frac{db(z)}{dz}h(z, b(z)) - \frac{da(z)}{dz}h(z, a(z)) + \int_{a(z)}^{b(z)} \frac{\partial}{\partial z}h(z, y)dy$$

Applying Leibniz rule:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(z - y, y)\, dy$$

Alternatively,

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z - x)\, dx$$

Convolution Formula

If X and Y are independent:

$$f_{XY}(x, y) = f_X(x)f_Y(y)$$

Thus,

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z - x)\, dx$$

This is the **Convolution Integral**.

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3. Special Cases

(i) Normal Random Variables

Let:

$$X \sim N(0, 1), \quad Y \sim N(0, 1)$$

Independent.

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}$$

Using convolution:

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \frac{1}{\sqrt{2\pi}} e^{-(z-x)^2/2} dx$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{x^2+(z-x)^2}{2}} dx$$

Simplifying exponent:

$$x^2 + (z - x)^2 = 2x^2 - 2zx + z^2$$

Complete square:

$$= 2\left(x - \frac{z}{2}\right)^2 + \frac{z^2}{2}$$

Thus,

$$f_Z(z) = \frac{1}{\sqrt{4\pi}} e^{-z^2/4}$$

Hence,

$$Z \sim N(0, 2)$$

(ii) Exponential Random Variables

Let:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y > 0$$

Using convolution:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

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$$= \lambda^2 e^{-\lambda z} \int_0^z dx$$

$$= \lambda^2 z e^{-\lambda z}, \quad z > 0$$

$$= 0, \quad \text{otherwise}$$

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4. Other Transformations

(a) $Z = X - Y$

$$F_Z(z) = P(X - Y \leqslant z)$$

$$= \iint_{x-y \leqslant z} f_{XY}(x, y)\, dx\, dy$$

Differentiate:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Procedure same as $Z = X + Y$.

(b) $Z = \frac{X}{Y}$

$$F_Z(z) = P\left(\frac{X}{Y} \leqslant z\right)$$

Region depends on sign of Y:

- $Y > 0$
- $Y < 0$

Integration region must be split accordingly.

(c) Example: $R = \sqrt{X^2 + Y^2}$

Find:

$$f_R(r)$$

Region corresponds to circular geometry as shown in lecture.

End of Lecture L11_12 Scribe

All derivations, limits, steps, and intermediate results strictly follow the lecture slides and are suitable for closed-notes examination revision.